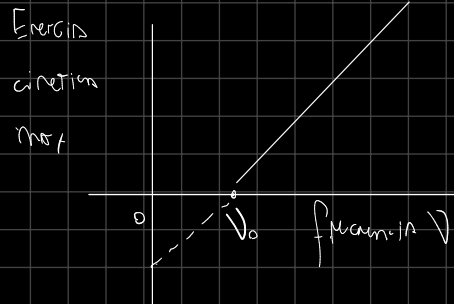
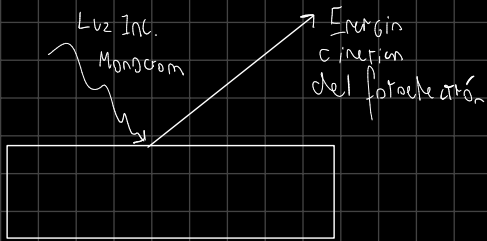


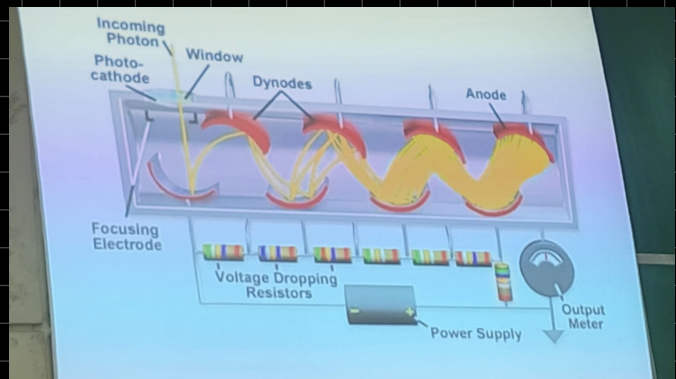
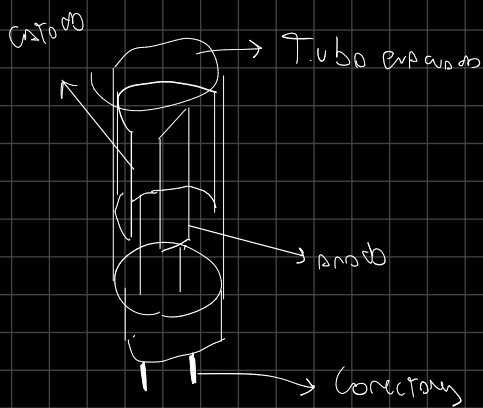
Obj:

Efecto Fotoelectrónico



$$E_{c \text{ máx}} = E_{\nu} - \Phi \Rightarrow V_0 = \frac{\Phi}{h}$$

$\downarrow$
 $h\nu$



Multiplica la corriente generada por un único fotón

Cuadro de Responsividad

$\frac{\text{mA}}{\text{W}}$ → Fotocorriente (I)
 $\downarrow$
 $I_{\nu} \cdot S_{\text{det}} = P_e$
 $\downarrow$
 $\left[\frac{\text{W}}{\text{cm}^2} \right]$

$\frac{\# \text{ fot}}{\text{cm}^2 \text{ s}}$
 Cantidad de fotones
 X Área X Tiempo

$$I_{\nu} = N_{\nu} h \nu$$

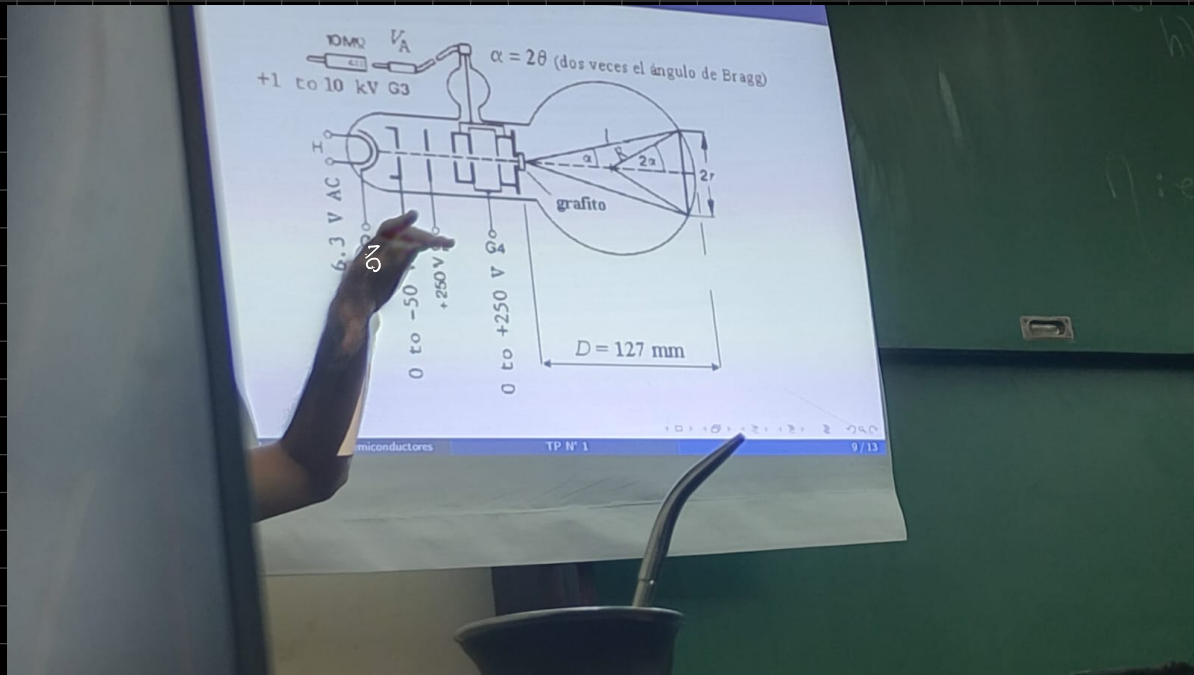
η = eficiencia cuántica

Cuántos fotones se necesitan para un fotoelectrón

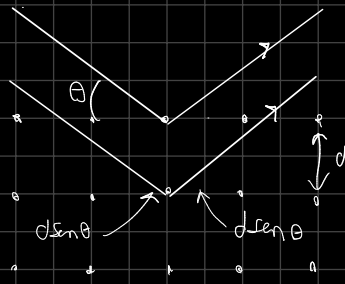
$$I = \underbrace{I_{\nu}}_{\text{Intensidad Incidente}} \underbrace{S_{\text{det}}}_{\text{superficie}} \underbrace{h \nu}_{\text{fotón}}$$

Cuadro de parámetros

$I_d [pA] \Rightarrow$ Corriente de oscuridad (piso de lo que puedo detectar)



Dispositivo basado en difracción de e^-



Para este
TP
 $n=1$

$$\leftarrow n\lambda_{dB} = 2d \sin(\theta)$$

Donde n es entero positivo, θ Ángulo de Bragg / en este dispositivo se cumple que $\alpha = 2\theta$

Practico

Ej: Schrödinger

$$\hat{H} \psi(x,t) = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + V(x,t) \psi(x,t) = j\hbar \frac{\partial \psi(x,t)}{\partial t}$$

Ind. del Tiempo

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2} + V(x) \psi(x) = E \psi(x)$$

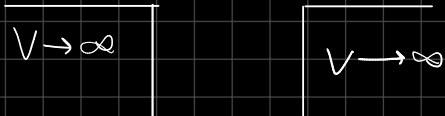
Energía

S: $V(x,t) = V(x) \rightarrow \psi(x,t) = \psi(x) \phi(t)$ $\phi(t) = \exp\left(-\frac{jEt}{\hbar}\right)$

Confinamiento

$\Rightarrow$ confinamiento

No confinamiento



$\psi(x) \Rightarrow$ es onda viajera

$\psi(x) \rightarrow$ onda estacionaria

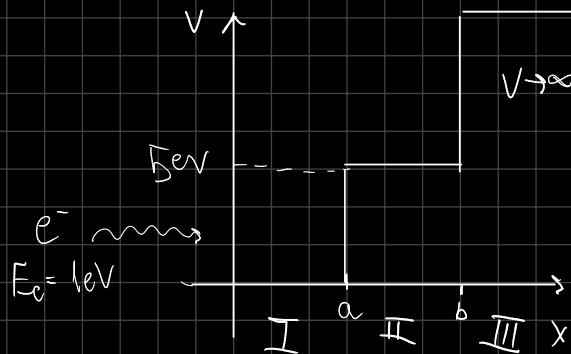
$$\psi(x) = A \exp(jkx) + B \exp(-jkx)$$

$$\psi(x) \propto \cos(kx) \propto \sin(kx)$$

Ej

a) plantear las expresiones de $\psi(x)$

b) Cuales es el numero de onda en cada Region? (K)



c) Calcular los coef. de Transmisión (T) y Reflexión (R)

a)

$$x < a \quad \psi(x) = A \exp(jk_1 x) + B \exp(-jk_1 x)$$

$$a < x < b \quad \psi(x) = C \exp(jk_2 x) + D \exp(-jk_2 x)$$

$$x > b \quad \psi(x) = \dots$$

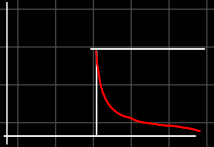
$$b) \quad k = \frac{p}{\hbar} \stackrel{E = \frac{1}{2} m v^2}{=} \sqrt{\frac{2mE}{\hbar^2}} \rightarrow \sqrt{\frac{2m(E-V)}{\hbar^2}} \begin{cases} K_1(V=0) = \sqrt{\frac{2mE}{\hbar^2}} \\ K_2(V=5\text{eV}) = \sqrt{\frac{2m(1\text{eV}-5\text{eV})}{\hbar^2}} \end{cases}$$

$$x < a \quad \psi_I(x) = A \exp(jK_1 x) + B \exp(-jK_1 x)$$

$$a < x < b \quad \psi_{II}(x) = C \exp(jK_2 x) + D \exp(-jK_2 x) = \underbrace{C \exp(-|K_2| x)}_{\text{onda evanescente}} + D \exp(|K_2| x)$$

$$x > b \quad \psi_{III}(x) = 0$$

onda evanescente

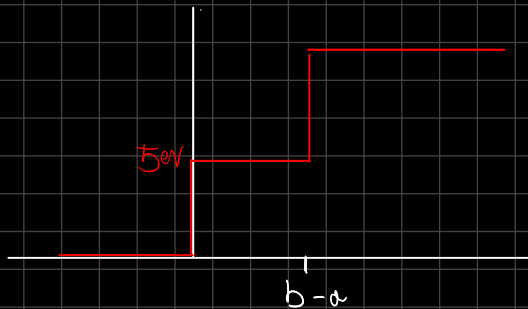


Condiciones de contorno / Frontera

$$\psi(b) = 0 \quad \psi(a) \stackrel{(i)}{\text{Continuo}} \quad \psi_I(a) = \psi_{II}(a)$$

$$\psi'(a) \stackrel{(ii)}{\text{Continuo}} \quad \psi'_I(a) = \psi'_{II}(a)$$

$$i) \quad A \exp(jK_1 a) + B \exp(-jK_1 a) = C \exp(-|K_2| a) + D \exp(|K_2| a)$$



$$\hat{x} = x - a$$

$$\psi_I(\hat{x}) = A \exp[jK_1(\hat{x}+a)] + B \exp[-jK_1(\hat{x}+a)]$$

$$\psi_I(\hat{x}) = \underbrace{A \exp[jK_1 a]}_{A'} \exp[jK_1 \hat{x}] + \underbrace{B \exp[-jK_1 a]}_{B'} \exp[-jK_1 \hat{x}]$$

me

$$\psi_I(0) = \psi_{II}(0) \rightarrow A' + B' = C' + D' \quad (w)$$

$$\psi'_I(0) = \psi'_{II}(0) \rightarrow jK_1 A' - jK_1 B' = jK_2 C' - jK_2 D' \Rightarrow K_1(A' - B') = K_2(C' - D') \quad (z)$$

$$\psi_{\text{I}}(b-a) = 0 \rightarrow C \exp[jk_2(b-a)] + D' \exp[-jk_2(b-a)] \quad (\gamma)$$

$$\stackrel{\alpha}{(\gamma)} \Rightarrow D = -C \exp[2jk_2(b-a)] \quad (\times)$$

$$(X) \text{ en } (z) \quad C = \frac{k_1}{k_2} \frac{(A-B)}{\{1 + \exp[2jk_2(b-a)]\}}$$

$$\swarrow \text{en } (w)$$

$$B = \frac{A \left(\frac{k_1}{k_2} g - 1 \right)}{\left(1 + \frac{k_1}{k_2} g \right)}$$

$$\text{Con } g = \frac{1 - \exp[2jk_2(b-a)]}{1 + \exp[2jk_2(b-a)]}$$

$$C = \frac{A \frac{k_1}{k_2} \left[1 - \left(\frac{k_1}{k_2} g - 1 \right) / \left(1 + \frac{k_1}{k_2} g \right) \right]}{1 + \exp[2jk_2(b-a)]}$$

$$D = \dots (\approx \approx \approx)$$

$$T = 1 - R$$

$$T_{(\text{I} \rightarrow \text{II})} = 0 \Rightarrow R_{(\text{I} \rightarrow \text{II})} = 1$$

$$R = \frac{U_1 B^* B}{U_1 A^* A} \rightarrow \frac{\text{refl. wave}}{\text{incidence}} = \frac{|B|^2}{|A|^2}$$

$$R = \frac{\left(\frac{k_2}{k_1} g - 1 \right)}{\left(\frac{k_2}{k_1} g + 1 \right)}$$

$$A = \frac{A'}{\exp(jk_1 a)}$$

$$B = \frac{B'}{\exp(-jk_1 a)}$$

$$C = \frac{C'}{\exp(jk_1 a)}$$

$$D = \frac{D'}{\exp(-jk_1 a)}$$